Polygon Structure-Guided Hyperspectral Image Classification With Single Sample for Strong Geometric Characteristics Scenes

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***Abstract*— Combining spectral and spatial information can significantly improve the classification performance of hyperspectral image (HSI). Currently, a lot of spectral–spatial HSI classification methods have been proposed. However, the task of HSI classification has remained challenging since the number of training samples is limited in real scenarios. In this article, we propose a novel HSI classification framework with single sample, in which the spectral self-similarity and spatial polygon structure information are fully combined to improve the classification performance. On the one hand, spectral self-similarity is used to expand training samples, which makes it possible to obtain sufficient samples with minimal cost. On the other hand, polygonal partition is introduced to acquire the geometrical structure of land covers in man-made environments. Specifically, the edge information of geometric objects is captured by polygonal partition, which can be utilized to constrain the spatial range of sample expansion and optimize the classification results. Experimental results on three real HSIs illustrate that the proposed method performs very well under small training sample size even when the number of samples is single per class.**

***Index Terms*— Hyperspectral image (HSI) classification, polygon structure, sample expansion, single sample, spectral similarity.**

I. Introduction

HYPERSPECTRAL images (HSIs) usually contain numerous spectral bands that can provide rich spectral information. Due to this advantage, hyperspectral images have been widely used in remote sensing researches, such as target detection [1], object classification [2], and spectral unmixing [3]. HSI classification is a hot research topic all the time since this technology can help people distinguish different land covers of interest, which is also beneficial for subsequent analysis and management decisions [4]–[6].

Over the past few years, various classification methods have been successfully applied for analyzing HSIs [7]–[9], which can be roughly divided into spectral-based classification methods and spectral–spatial classification methods. Spectral-based classification methods identify different objects based on the spectral information, including support vector machine (SVM) [10] and multinomial logistic regression (MLR) [11]. With the development of deep learning, deep models have also been introduced to classify HSIs and proved effective, e.g., stacked autoencoders (SAEs) [12], deep belief networks (DBNs) [13], convolutional neural networks (CNNs) [14], and recurrent neural networks (RNNs) [15]. However, the performance of most existing methods highly relies on the training sample size. Unfortunately, acquiring sufficient labeled samples is usually difficult and costly in many applications.

In order to solve the small labeled training sample size problem, many spectral–spatial features extraction methods are proposed, which can construct discriminant features to increase the separability of different ground objects. For instance, a series of morphological profile-based methods are proposed in [16]–[19], which can obtain the spatial features through morphological opening and closing operations with different structuring elements. In [20], Kang *et al.* developed edge-preserving filtering with different parameters on HSI, which can capture spatial information of objects with different scales. In addition, low-rank representation (LRR) is able to construct data relationships for HSI classification and has been proved to be an effective method for HSI classification [21]–[24]. In [23], Wang *et al.* proposed a structure-regularized LRR framework that fully utilizes local correlation among neighboring pixels to obtain the global and local structure features. To eliminate the influence of noise, Duan *et al.* [25] introduced a multiscale total variation (MSTV) method to extract structural features.

In addition, an alternative method is to generate abundant training set, such as semisupervised learning (SSL) [26], active learning (AL) [27], and collaborative learning (CL). The SSL strategy aims at selecting valuable unlabeled samples and assigning pseudo label for these samples. For instance, a graph-based SSL method for HSI classification is

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proposed [28], which utilizes the unlabeled samples information and the integration of contextual details. Li *et al.* [29] proposed an SSL algorithm that exploits both the spatial contextual information and the spectral information to enrich training set. Unlabeled samples are selected by means of an active-selection strategy based on the entropy of samples in this method. Tan *et al.* [30] used breaking ties technology and spatial neighborhood information to find the helpful unlabeled samples in the semisupervised process. Then, they utilize MLR and k-nearest neighbor (KNN) to estimate the labels of unmarked samples. Furthermore, the AL approaches aim at establishing efficient training set by iteratively increasing the model performance through sampling [31]. Di and Crawford [32] embedded data regularization into the AL paradigm and proposed a novel sequential coregularizer for HSI classification. Samat *et al.* [33] combined the edge gradient information with the traditional AL framework. Zhou *et al.* [34] utilized the 3-D redundant wavelet transform that is insensitive to additive noise to produce multiple views for AL framework. Cao *et al.* [35] extended the AL method to the graph-based CNN model for HSI classification, through selecting the most informative samples actively to fine-tune CNN and improve classification accuracy. Besides, CL methods acquire richer samples information through a combination of SSL and AL [36]. Sun *et al.* [36] proposed a novel random walker-based CL framework, which can obtain more high confidence samples. SSL and AL are also combined to form a collaborative labeling procedure by classifiers and manual labeling in [37]. Moreover, Wang *et al.* [38] used CL to discover the representative and discriminative information from the unlabeled pixels.

Although many existing classification methods can acquire satisfactory results when training samples are relatively limited, the classification performance is still unsatisfactory when the number of training samples is single per class. To overcome this problem, this article proposes a polygon structure-guided HSI classification framework with single sample. The main motivation of this framework is that the polygon structure is appropriate for analyzing scenes with strong geometric characteristics [39], such as crop or urban scene. Taking farmland as a case study, polygons are able to represent the boundaries of most fields. Besides, crops in the same field have a great probability of belonging to the same category. By combining the spatial information of the polygon structure and the spectral information, the classification performance can be substantially improved. Specifically, the spectral dimensionality of the original HSI is reduced with minimum noise fraction (MNF). In order to make the edge information of land covers clearer, image enhancement is preformed on the first three components of dimension reduced image. The enhanced image is segmented so as to generate a polygon structural map. Then, the partition result is used as spatial constraint and combined with spectral features for searching more training samples. Finally, a classification map is obtained by performing a pixelwise classifier SVM on the dimension reduced image and further optimized by polygon structural maps. To summarize, the main contributions of this article are listed as follows.

1) A polygonal partition method is explored for the first time to solve the small sample size problem. The spatial structure and spectral similarity are fully utilized to expand the number of the training set.
2) A novel HSI classification method with single sample is proposed. Experiments on crop scenes demonstrate that the classification maps are more consistent with human perception. The edges of different land covers are well aligned with the original images.
3) Experiments are carried out on three HSIs of crop scenes to confirm the effectiveness and advantage of the proposed method. Experimental results indicate that the proposed method greatly outperforms other state-of-the-art approaches even when the number of training samples is single per class.

The remainder of this article is organized as follows. The polygonal partition algorithms are reviewed in Section II. Section III presents the details of the proposed classification framework. Section IV reports the experimental results and analyses. Finally, conclusions are provided in Section V.

II. Polygonal Partition

Superpixel segmentation methods can divide the image into a set of small regions with internal homogeneity, which is commonly used for HSI classification [40], [41]. For example, superpixel map can be used as a spatial constraint to optimize initial classification results with majority voting. In addition to superpixel segmentation, recent studies show that polygonal partition has become a powerful tool in computer vision. Similar to superpixels, polygons are used to represent internal homogeneous regions. Comparing with traditional pixel-based superpixel segmentation, the significant distinction is that the spatial coordinates of the image encoded by polygonal partition and superpixel segmentation are continuous and discrete, respectively, [42]. Moreover, some researchers [39], [43], [44] also demonstrate that polygonal partition performs better to capture and analyze the geometric signatures of images, especially in man-made environments.

Existing polygonal partition algorithms mainly include two categories: superpixel boundary-based algorithms [44] and line segment-based algorithms [39], [43]. For the superpixel boundary-based algorithms, image is partitioned into superpixels and then vectorizing superpixel boundaries to get polygons. For example, Achanta and Ssstrunk [44] proposed a simple noniterative clustering algorithm (SNIC) to generate a superpixel map. Considering that adjacent superpixels share the same polygon edges, the superpixel boundaries are modified to transform superpixels into polygons. The line segment-based algorithms detect the line segments and then connect them to generate polygons. Duan and Lafarge [43] segmented the image into convex polygons by building a Voronoi diagram that conforms to predetected line segments. In [39], a kinetic approach named kinetic polygonal partitioning of images (KIPPI) is proposed, which can obtain polygons with different sizes. The core idea of this approach is extending predetected line segments extracted by the line segment detector (LSD) [45] until intersection of these line segments.

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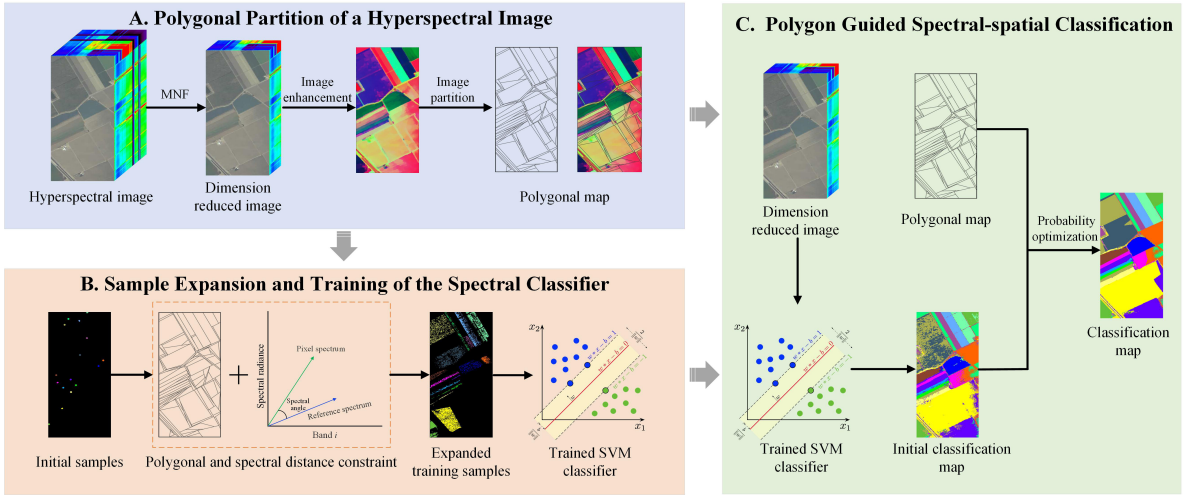

Fig. 1. Flowchart of the proposed classification framework. It mainly consists of three stages. First, the dimension of input image is reduced to get a polygonal map. Then, using the polygonal map to enlarge the training set for training the SVM classifier. Finally, the trained classifier is performed on the dimension reduced image to get a classification result that is further optimized by the polygonal map.

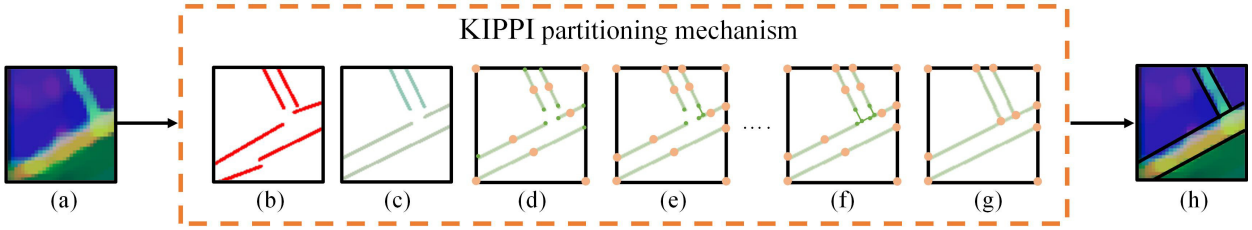

Fig. 2. Illustration of KIPPI partitioning mechanism. (a) Input image. (b) Original line segments. (c) Optimized line segments. (d) Initialization of graph. (e) and (f) Extension of line segments, as extremities of primitives extend (green dots), new vertices and edges are insert to enrich planar graph. (g) Finalization, the planar graph is simplified by removing unnecessary vertices. (h) Partition result.

III. Proposed Method

The proposed classification method is introduced in this section, and its schematic is presented in Fig. 1. This method is composed of three main steps: polygonal partition, sample expansion and training of the spectral classifier, and polygon-guided spectral–spatial classification. The detailed steps are shown as follows.

A. Polygonal Partition of an HSI

In order to obtain more accurate polygonal partition result, the preprocessing is necessary for the original HSI $\mathbf{O} \in \mathbb{R}^{M\times N\times B}$, $M \times N$ is the spatial size of $\mathbf{O}$, and B is the original spectral band number. First, the spectral dimension of HSI is reduced to K by MNF, which is a universal and effective tool to reduce the dimension of HSI while eliminating noise in the image. Besides, the first three principal components of the HSI are selected and the contrast of the false-color image is enhanced with histogram equalization. Then, a total variation-based method [46], which can decompose the input image into structure and texture components, is adopted to enhance the main spatial structures, which can improve the identifiability of boundaries. Note that the aforementioned two steps are performed on the first three components of the dimension reduced HSI. Finally, the KIPPI [39] algorithm is applied to the enhanced image. Fig. 2 shows the partitioning mechanism of KIPPI, which mainly includes three steps.

1) Line Segments Detection: At first, line segments are detected in the enhanced image [45], as shown in Fig. 2(b). In order to understand the next steps of the algorithm, $\beta = (\beta_1, \ldots, \beta_n)$ is used to represent the initial orientation (the angle with level line) of these line segments, where n is the number of line segments.

2) Global Regularization: In order to correct the imprecisions of line segments, this step is adopted to reorient and realign line segments based on the characters (parallelism, orthogonality, and collinearity) of man-made objects, as shown in Fig. 2(c). It should be noted that this step is optional and can be expressed by minimizing the energy function

$$U(x) = (1-\lambda)H(x) + \lambda V(x). \tag{1}$$

In the line segments reorientation task, $\mathbf{x} = (x_1, \ldots, x_n)$, $x_i \in [-\theta_{\max}, \theta_{\max}]$ represents the amount to be added to the initial orientation β_i of the line segment i. The weighting parameter $\lambda \in [0, 1]$, with the default value 0.8. The data term $H(x)$ is designed as

$$H(x) = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{x_i}{\theta_{\max}}\right)^2. \tag{2}$$

$H(x)$ is used to restrain strong angle deviations. Besides, pairwise potential $V(x)$ can turn approximately parallel or orthogonal line segment pairs into completely parallel or

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orthogonal, which is expressed by

$$V(x) = \frac{1}{\sum_{i=1}^{n}\sum_{j>i}\mu_{ij}} \sum_{i=1}^{n}\sum_{j>i}\mu_{ij}\frac{\left|\theta_{ij} - x_i + x_j\right|}{4\theta_{\max}} \tag{3}$$

$$\theta_{ij} = \begin{cases} \theta_{ij} = \alpha_{ij}(\mathrm{mod}\pi), \\ \quad \text{if } \alpha_{ij} \in \left[-\frac{\pi}{4}, \frac{\pi}{4}\right] \cup \left[\frac{3\pi}{4}, \frac{5\pi}{4}\right] \\ \theta_{ij} = \alpha_{ij} - \frac{\pi}{2}(\mathrm{mod}\pi), \\ \quad \text{otherwise} \end{cases} \tag{4}$$

θ_{ij} is the deviation of the relative angle α_{ij} between line segment i and j. If line segments i and j are spatially adjacent and $|\theta_{ij}| < 2\theta_{\max}$, $\mu_{ij} = 1$. If not, $\mu_{ij} = 0$.

In the line segments realigning task, $H(x)$ is redefined as $H(x) = \sum_{i=1}^{n}(x_i/d_{\max})^2$ and $V(x)$ is redefined as $V(x) = \sum_{i=1}^{n}\sum_{j>i}\mu_{ij}^{*}(|d_{ij} - x_i + x_j|/4d_{\max})$. $x_i \in [-d_{\max}, d_{\max}]$ represents the offset performed on line segment i along the orthogonal direction. d_{ij} represents the distance between parallel line segments i and j. When line segments i and j are parallel, $\mu_{ij} = 1$ and $d_{ij} < 2d_{\max}$, $\mu_{ij}^{*} = 1$. Otherwise, $\mu_{ij}^{*} = 0$.

3) Kinetic Partition: This step is achieved by iteratively updating dynamic planar graph $G_t = (V_t, E_t)$, as shown in Fig. 2(d)–(g). V_t and E_t are the sequence of vertices and edges at time t, respectively. To describe this kinetic framework, some variables are defined where the certificate function $C_i(t)$ is expressed as follows:

$$C_i(t) = \prod_{\substack{j=1 \\ j\neq i}}^{N} Pr_{i,j}(t). \tag{5}$$

In the kinetic framework, primitive corresponds to half line segment, and N is the number of primitives. The predicate function $Pr_{i,j}(t)$ returns 0 when primitive i collides with j and 1 otherwise. Moreover, priority queue is created by computing and sorting all the times for that $C_i(t) = 0$.

The initialization settings of graph G include inserting some points as vertices, such as the four corner points of the image, the midpoint of each line segment, and the intersection of two line segments. Then, edges are set between the four successive corner points, possible intersection points, and the midpoints of their respective line segments. With the extension of the line segment, new vertices and edges are added to graph G to keep it planar. The process of iterative updating is finished until priority queue is null. Then, redundant vertices from G are deleted to simplify the partition result. In this work, we denote the polygonal partition result obtained by KIPPI as $\mathbf{P}$.

B. Sample Expansion and Training of the Spectral Classifier

For HSI classification task, spectral information is very significant. In addition, utilizing spatial information is able to further improve the classification accuracy, which can preserve the structural feature of the objects. Therefore, spectral–spatial information can be jointly utilized to enlarge the training sample set. In this section, a spectral–spatial sample expansion method guided by the polygonal map $\mathbf{P}$ is introduced. This method adopts spectral angle distance and polygon structure to search similar samples inside and outside the polygon, respectively, and more details of this method are shown as follows.

1) Sample Expansion Inside the Polygon.: In this step, similar samples are searched inside the polygon $\mathbf{P}_c$ in which the initial single training sample s is located. Assuming that $\mathbf{P}_c$ contains p pixels, s_i $(i = 1, \ldots, p-1)$ is its ith pixel. Then, the spectral angle distance D_i is employed to estimate the similarity of s and s_i, which is calculated by

$$D_i = \cos^{-1}\left(\frac{s^T s_i}{\|s\| \cdot \|s_i\|}\right). \tag{6}$$

When $D_i < \eta$, the corresponding pixel s_i is considered to have high spectral similarity with initial sample s and labeled as the same class with s. The threshold η is set to the value of bin with the largest number in the histogram of spectral distance D_i $(i = 1, \ldots, p-1)$.

The motivation of this step is that pixels in the same polygon have high spatial correlation and can be divided into two parts, with the same class of initial sample s or not. Besides, polygon also represents the internal homogeneous region, which means that most pixels in $\mathbf{P}_c$ are similar to s. Therefore, according to the statistical characteristics of spectral distance D_i, the histogram is used to determine the parameter η.

2) Samples Expansion Among the Polygons: Since the spatial distribution of objects that belonging to the same class is sometimes not adjacent, it is necessary to expand samples among the polygons. In this step, similar polygons are found by spectral distance constraint at first, and then, samples are searched within these polygons.

Assuming that $\mathbf{P}$ contains $\hat{p}$ polygons, $\mathbf{q} = (q_1, \ldots, q_{\hat{p}-1})$ is the spectrum of these polygons except $\mathbf{P}_c$, which is calculated by the average spectrum of pixels belonging to the same polygon. The spectral distance between s and $\mathbf{q}$ is also obtained by (6). Then, these spectral distances are reordered in ascending order. The polygons corresponding to top $\hat{p}/\hat{c}$ spectral distances are considered to have high similarity with the initial sample and denoted as $\mathbf{P}_q$, where $\hat{c}$ represents the number of ground object categories. Next, samples are expanded in the similar polygons $\mathbf{P}_q$. In order to avoid the introduction of mislabeled samples, double spectral distance constraints are proposed. In more detail, two spectral distances are calculated: 1) D_q is the spectral distance between s and pixels in $\mathbf{P}_q$ and D_q^{*} is the spectral distance between s^{*} and pixels in $\mathbf{P}_q$, where s^{*} represents the average spectra of samples obtained through sample expansion inside the polygon $\mathbf{P}_c$. Similar to sample expansion inside the polygon, a threshold is adopted to expand samples in this step. Specifically, when $D_q < \mu$ and $D_q^{*} < \mu$, the corresponding pixels are added to the training sample set. It should be noted that as the value of μ increases, more samples can be searched, but more mislabeled samples are also introduced. In our experiments, μ is set to 0.01 typically.

The spectral distance thresholding criteria are different in the above two-step sample expansion processes since the distributions of similar pixels are different. Through the above expansion processes, a large number of labeled samples can be searched to train the SVM classifier. Afterward, the SVM

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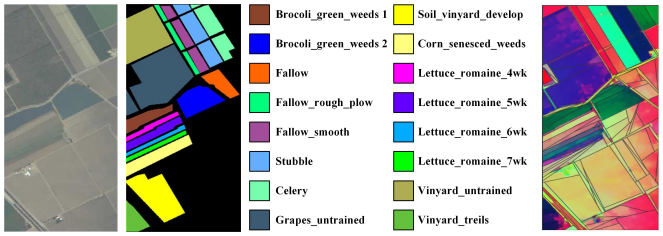

Fig. 3. Salinas dataset. (a) Color composite. (b) Ground truth. (c) Labels. (d) Polygonal map.

classifier is conducted on the dimension reduced image $\mathbf{I} \in \mathbb{R}^{M\times N\times K}$ to obtain the initial classification map. It should be mentioned that other classifiers can also be considered in the proposed framework.

C. Polygon-Guided Spectral–Spatial Classification

Since the classification process has not considered the spatial information adequately, the initial classification result estimated by SVM contains mislabeled noisy pixels inevitable and not align with the boundaries of land covers. Similar to the postprocessing method proposed by Tarabalka [47]–[49], the polygonal map $\mathbf{P}$ is adopted to optimize the classification result to improve the classification accuracy in this work. Let $\mathbf{P}_h$ represent the hth polygon in $\mathbf{P}$, $\mathbf{L}_h$ denote the initial labels for pixels in $\mathbf{P}_h$, and L_w $(w = 1, \ldots, \hat{c})$ denote the number of pixels labeled as the wth classes in $\mathbf{P}_h$. Then, the percentage of pixels belonging to $\mathbf{P}_h$ with label w can be calculated by

$$P_h^w = \left(\frac{L_w}{\sum_{w=1}^{\hat{c}} L_w}\right). \tag{7}$$

Then, the final label of $\mathbf{P}_h$ is obtained by

$$\mathbf{L}_\mathbf{h}^* = \begin{cases} w, & \text{if } P_h^w > 0.5 \\ \mathbf{L}_h, & \text{otherwise.} \end{cases} \tag{8}$$

IV. Experiments

A. Datasets

To verify the effectiveness of the proposed method, three hyperspectral datasets contain a wide variety of crops that are employed here. They are briefly described as follows.

1) *Salinas*[1]*:* This dataset is a representative crop scene containing sixteen classes of interest and captured by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor over Salinas Valley, California. After removing useless bands, 204 bands are retained for classification. Besides, the spatial size is 512 × 217 and the spatial resolution is 3.7 m per pixel. The ground truth includes 16 different classes and 54 129 labeled samples. Fig. 3 shows the three-band color composite, ground truth, and partition map.

[1]http://www.ehu.eus/ccwintco/index.php/Hyperspectral_Remote_Sensing_Scenes

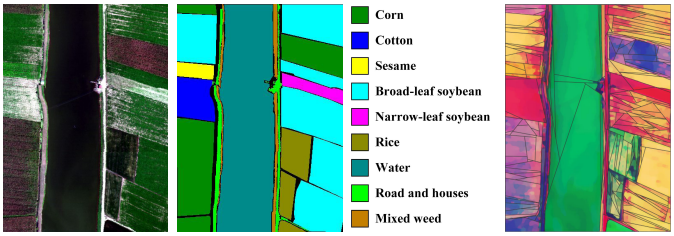

Fig. 4. WHU-Hi-LongKou dataset. (a) Color composite. (b) Ground truth. (c) Labels. (d) Polygonal map.

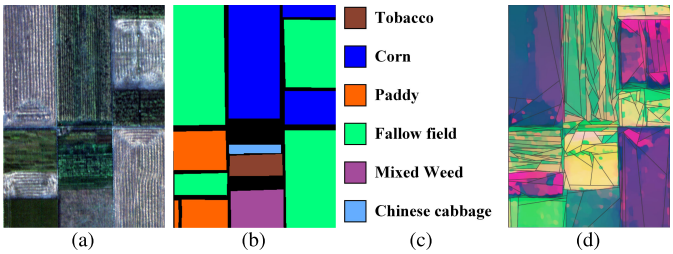

Fig. 5. Yunyuan dataset. (a) Color composite. (b) Ground truth. (c) Labels. (d) Polygonal map.

2) *WHU-Hi-LongKou*[2]*:* This dataset is published in [50]. The imaging location is Longkou Town, Hubei province, China. This image was captured by the Nano-Hyperspec equipped on an unmanned aerial vehicle (UAV) platform. This image contains 270 bands of size 550 × 400 and the spatial resolution is about 0.463 m. Its false-color image, ground truth, and polygonal map are presented in Fig. 4(a), (b), and (d), respectively.
3) *Yunyuan:* Yunyuan data are captured in the experimental field of Hunan Agricultural University, Changsha, China, on November 13, 2020. The airborne imaging spectrometer is Headwall Nano-Hyperspec, which is equipped with Da Jiang Innovations (DJI) Matrice 600 Pro UAV platform. These original data consist of 270 spectral channels ranging from 400 to 1000 nm. After removing the invalid bands, 250 bands are reserved for classification. The spatial size is 385 × 530 with almost 0.247-m spatial resolution. Fig. 5 shows the RGB composite, ground truth, and polygonal map. It is worth mentioning that necessary preprocessing (e.g., radiometric calibration and geometric correction) is adopted to these data with HyperSpec software supplied by the imaging spectrometer manufacturer. Moreover, the ground truth is obtained through fieldwork.

B. Experimental Setup

1) Evaluation Metrics: Three commonly employed objective metrics, namely overall accuracy (OA), average accuracy (AA), and Kappa coefficient (KA), are used to quantitatively measure the classification efficiency of all tested methods. Specifically, OA measures the proportion of correctly classified pixels in the test samples. AA is the mean of the percentage of pixels accurately defined for each class. KA is used to measure the proportion of identified pixels that are adjusted by the amount of agreements that will be predicted by chance.

[2]http://rsidea.whu.edu.cn/resource_WHUHi_sharing.htm

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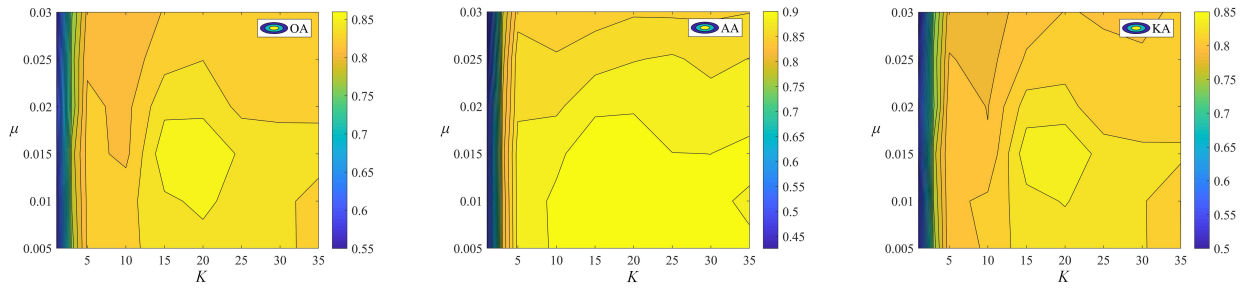
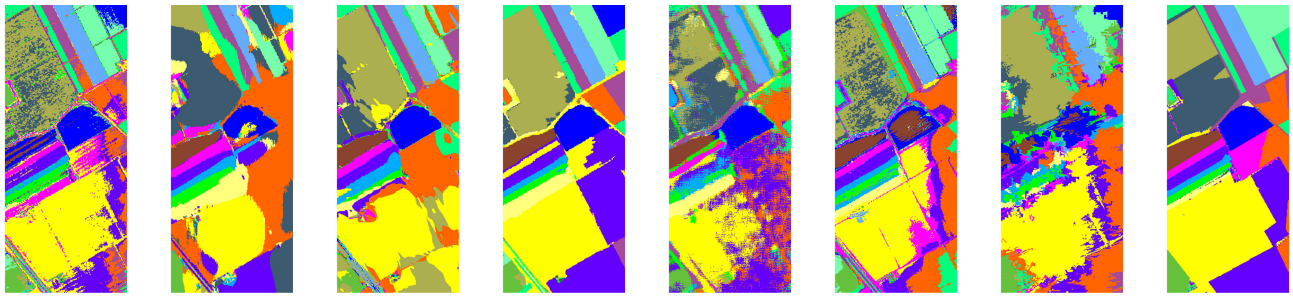


Fig. 6. Influence of band number K and the threshold of spectral distance μ to the classification performance of the proposed method (Salinas dataset). (a) OA. (b) AA. (c) KA.



(a) OA=70.40% (b) OA=67.02% (c) OA=73.51% (d) OA=74.99% (e) OA=66.93% (f) OA=67.71% (g) OA=61.57% (h) OA=85.67%

Fig. 7. Classification maps of all studied methods on the Salinas dataset. (a) SVM [10]. (b) PCA-EPFs [20]. (c) MSTV [25]. (d) PKCRCAWG [51]. (e) GTR [52]. (f) RPNet [53]. (g) SCMK [40]. (h) Our method.

2) Comparison Methods: In this work, seven state-of-the-art HSI classification methods are adopted to compare with our method, including SVM [10], principal component analysis-based edge-preserving features method (PCA-EPFs) [20], MSTV-based method [25], probabilistic-kernel collaborative representation with adaptive weighted graph-based method (PKCRCAWG) [51], generalized tensor regression-based method (GTR) [52], random patches network-based method (RPNET) [53], and superpixel-based multiple kernels method (SCMK) [40].

3) Parameter Settings: For the above studied methods, parameters are set to default values according to relevant articles. Besides, the proposed method includes two parameters that need to be determined: the band number of the dimension reduced image K and the threshold of spectral distance μ. The following experiments will illustrate the basis for setting these parameters.

C. Parameter Analysis

In this section, the Salinas dataset is adopted to analyze the effect of aforementioned parameters on classification result and initialize them. The variation interval of parameters K and μ is set to be [1, 35] and [0.005, 0.03], respectively. Fig. 6 shows how the OA, AA, and KA vary with K and μ. Obviously, when $K = 20$ and $\mu \in [0.01, 0.015]$, above accuracy indexes reach the maximum value. It can also be seen that the proposed method is influenced by the preserved band number K apparently. When K is relatively small, all accuracy indexes are reduced. When K are relatively large, OA and KA tend to decrease and AA is stable. The changing trend of spectral distance threshold μ is similar to K, and the value should set in an appropriate range to control the quantity and quality of samples. Considering the computation cost and classification performance, K is set to 20 and μ is set to 0.01. The following experiments on the three datasets demonstrate that the default parameter setting is able to obtain satisfactory classification results.

D. Classification Results from a Single Training Sample

The classification results of various classification methods on three HSI datasets (i.e., Salinas, Wuhan University (WHU)-Hi-LongKou, and Yunyuan) are presented in this section. For each method, the experimental result is calculated as the average values of the ten results on different randomly selected samples. It should be noted that all methods are tested with single training sample per class in this section.

The experiment is first conducted on the Salinas dataset, and Table I presents the quantitative results of all considered approaches. The corresponding classification maps are shown in Fig. 7. As shown in Fig. 7, the classification map obtained by spectral classifier SVM exists noisy labels since this method only utilizes spectral features of HSI. The RPNet and GTR methods have improved this noise-like misclassification phenomenon. Besides, the PCA-EPFs, MSTV, and PKCR-CAWG methods that combine spatial information and spectral information can obtain more homogenous classification maps but produce mislabeled pixels. The SCMK method based on superpixel segmentation can also acquire smooth map

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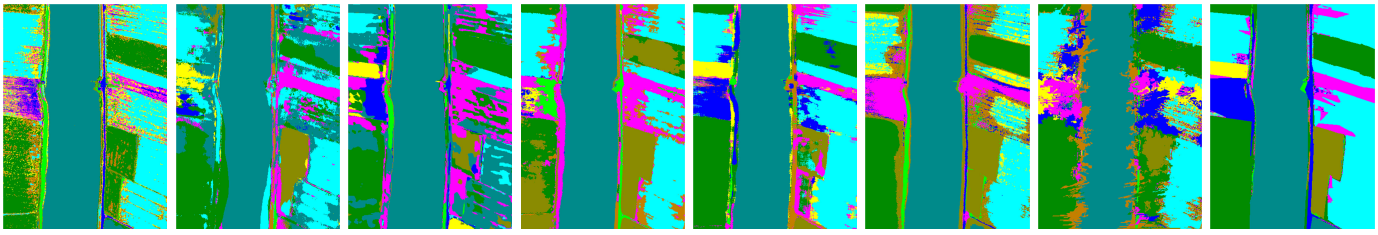
TABLE I

CLASSIFICATION PERFORMANCE OF ALL STUDIED APPROACHES FOR SALINAS DATASET WITH SINGLE SAMPLE, INCLUDING SVM [10], PCA-EPFS [20], MSTV [25], PKCRCAWG [51], GTR [52], RPNET [53], SCMK [40], AND OUR METHOD

Class name	Classification accuracies of different approaches (in %)							
	SVM	PCA-EPFs	MSTV	PKCRCAWG	GTR	RPNet	SCMK	Our method
Broccoli_green_weeds1	71.82±16.84	99.84±0.33	98.41±4.95	85.46±33.24	95.07±5.90	71.90±24.01	51.82±22.31	**99.50±0.05**
Broccoli_green_weeds2	94.12±9.62	97.93±3.45	93.44±16.70	89.99±31.62	98.76±1.39	49.51±23.84	55.99±27.15	**99.96±0.12**
Fallow	63.09±24.06	49.11±31.08	63.06±23.77	69.47±47.95	65.68±16.17	58.36±16.39	53.12±18.05	**97.67±2.59**
Fallow_rough_plow	95.52±3.42	55.99±17.52	82.46±6.62	88.55±31.12	46.78±17.56	**99.12±1.29**	85.21±9.31	93.85±2.43
Fallow_smooth	**89.38±10.53**	88.11±7.48	86.64±10.63	83.54±21.02	67.64±17.18	69.00±18.78	43.62±21.71	85.34±13.93
Stubble	99.97±0.04	99.80±0.36	99.41±0.50	**100.00±0.00**	99.59±0.81	96.62±2.52	84.08±5.64	**100.00±0.00**
Celery	89.84±5.29	92.77±8.89	99.39±1.26	92.92±4.99	99.73±0.43	75.79±20.85	78.52±10.36	**99.99±0.03**
Grapes_untrained	63.72±9.40	82.52±17.35	74.58±15.68	54.80±21.41	68.42±23.80	53.85±23.43	32.96±22.54	**93.04±12.02**
Soil_vinyary_develop	91.15±8.50	63.52±17.76	84.83±13.07	**98.25±4.86**	93.09±4.95	86.79±27.86	95.55±1.62	82.70±20.44
Corn_senesced_weeds	66.51±20.97	87.08±18.49	**94.58±5.90**	81.53±32.90	82.45±15.74	53.10±14.76	33.77±14.96	91.90±10.43
Lettuce_romaine_4wk	59.53±20.01	52.83±19.61	94.98±3.22	71.43±26.95	**96.30±2.71**	90.40±20.41	50.57±24.27	86.38±7.81
Lettuce_romaine_5wk	68.58±8.21	43.04±12.32	71.64±18.58	**99.13±0.91**	47.24±19.99	94.53±12.16	72.00±16.50	88.34±17.23
Lettuce_romaine_6wk	58.05±12.94	47.66±18.27	63.35±16.33	85.23±14.14	26.25±15.83	**98.58±0.89**	59.84±20.65	95.75±5.34
Lettuce_romaine_7wk	65.88±17.26	71.25±19.23	71.03±24.59	**98.38±2.09**	43.27±23.98	89.64±5.08	73.90±19.01	81.14±19.80
Vinyard_untrained	38.81±8.11	**77.65±22.64**	51.17±13.33	38.82±7.19	46.31±9.73	44.88±26.26	73.52±22.41	71.21±19.64
Vinyard_trellis	85.09±26.86	99.76±0.24	99.71±0.30	**100.00±0.00**	98.75±3.41	73.97±20.63	71.77±5.75	99.47±0.34
OA	70.40±2.50	67.02±5.41	73.51±5.36	74.99±5.14	66.93±8.27	67.71±4.37	61.57±3.73	**85.67±6.11**
AA	75.07±2.09	75.55±3.67	83.04±2.92	83.59±7.96	73.46±4.94	75.38±3.08	63.51±3.84	**91.64±1.88**
Kappa	67.29±2.67	64.10±5.79	70.87±5.65	72.32±5.65	63.80±8.84	64.32±4.69	57.77±3.88	**83.93±6.89**



(a) OA=65.62% (b) OA=56.41% (c) OA=56.24% (d) OA=68.55% (e) OA=75.86% (f) OA=70.97% (g) OA=65.31% (h) OA=85.20%

Fig. 8. Classification maps of all studied methods on the WHU-Hi-LongKou dataset. (a) SVM [10]. (b) PCA-EPFs [20]. (c) MSTV [25]. (d) PKCRCAWG [51]. (e) GTR [52]. (f) RPNet [53]. (g) SCMK [40]. (h) Our method.

but not performs well around the edges of different objects. Although the PKCRCAWG method provides more accurate results near the boundary, a large area has been mislabeled. In particular, "Grapes_untrained" is misclassified as "Vinyard_untrained"Comparing with these methods, the proposed method acquires a more accurate classification map, which indicates that sample expansion is able to improve the classification accuracy. Moreover, our method reaches the highest value according to Table I, with OA = 85.67%, AA = 91.64%, and KA = 83.93%.

The second experiment is performed on the WHU-Hi-LongKou dataset. Fig. 8 shows the classification maps obtained by all aforementioned approaches. Obviously, "noise" still exists in the SVM classification result. By contrast, the GTR and RPNet methods achieve better performance. All methods also fail to acquire accurately ground object boundaries and the homogeneity of each field cannot be guaranteed. This is mainly because few training samples are available for training the classifier. As shown in Fig. 8, the ultimate classification map obtained by the proposed method is more aligned with ground object boundaries and smoother in regions of "Corn" and "Broad-leaf soybean" since our method makes full use of the polygon structural information to optimize initial classification map. However, it is challenging to distinguish the object that is tiny and not with polygonal shape, such as "Mixed weed." The classification results of different methods are listed in Table II. As shown in Table II, it can be observed that our method is suitable to identify the land covers with large coverage, such as "Corn" and "Cotton" which also demonstrates the effectiveness of spatial probability optimization. Besides, according to three quantitative indicators, our method still provides a competitive evaluation index and outperforms other studied methods, with OA = 85.20%, AA = 78.06%, and KA = 81.15%.

The third experiment is conducted on the Yunyuan dataset. The corresponding classification maps are shown in Fig. 9. It can be seen that the majority of methods yield noise-liked misclassification maps due to the low spectral discrimination of various plants. For instance, the "Fallow field" is misclassified as "Mixed Weed" in all methods. This situation is inevitable due to the intraclass spectrum variability. In contrast, the labels of pixels in the same object are more accurate in our method, which is due to that the spatial optimization eliminates

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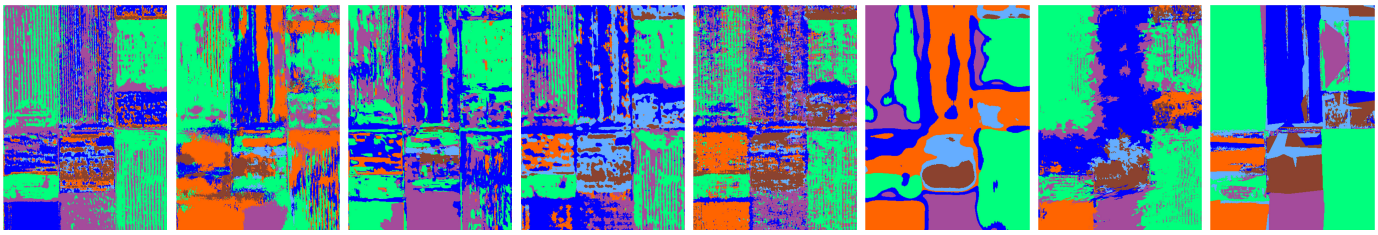
TABLE II

CLASSIFICATION PERFORMANCE OF ALL STUDIED APPROACHES FOR WHU-HI-LONGKOU DATASET WITH SINGLE SAMPLE, INCLUDING SVM [10], PCA-EPF [20], MSTV [25], PKCRCAWG [51], GTR [52], RPNET [53], SCMK [40], AND OUR METHOD

Class name	Classification accuracies of different approaches (in %)							
	SVM	PCA-EPFs	MSTV	PKCRCAWG	GTR	RPNet	SCMK	Our method
Corn	81.19±14.22	55.92±18.66	49.84±11.12	78.21±18.21	84.14±7.53	91.47±4.23	83.52±8.46	**92.77±10.02**
Cotton	33.41±18.06	**80.42±14.55**	67.29±11.04	65.42±18.72	38.71±21.02	38.63±17.58	52.72±21.09	29.51±15.61
Sesame	5.86±2.16	67.75±22.00	60.95±19.18	5.88±6.38	49.34±23.16	60.05±19.85	47.39±10.81	**81.58±30.23**
Broad-leaf soybean	80.25±25.16	78.94±7.74	78.14±10.95	90.77±9.19	92.69±5.08	35.48±22.18	39.25±16.71	**99.27±1.58**
Narrow-leaf soybean	17.51±6.43	47.38±28.34	37.89±32.54	17.67±7.57	24.06±12.58	**62.09±19.15**	44.26±14.79	45.23±30.35
Rice	65.50±17.26	83.06±14.91	76.77± 20.75	73.55±26.34	94.18±8.00	81.76±26.28	46.67±16.01	**97.89±1.26**
Water	98.86±0.43	55.58±12.61	60.41±7.23	98.59±0.55	98.52±2.26	**99.65±0.78**	93.13±3.29	97.74±0.80
Road and houses	49.66±17.34	**87.97±14.47**	78.45±17.84	76.42±22.91	45.79±23.41	53.59±15.72	29.86±10.53	85.43±20.09
Mixed weed	12.64±4.74	78.52±15.55	**85.59±15.55**	13.45±13.79	33.21±24.09	61.03±20.82	41.34±15.07	73.08±30.73
OA	65.62±8.45	56.41±5.90	56.24±2.89	68.55±5.63	75.86±11.48	70.97±7.61	65.31±4.70	**85.20±5.24**
AA	49.43±3.69	70.62±4.03	66.15±3.07	57.77±3.32	62.29±5.09	64.86±5.96	53.13±3.36	**78.06±7.69**
Kappa	57.80±9.47	41.11±8.94	42.25±3.48	60.94±5.97	69.88±12.80	64.65±8.54	57.51±5.11	**81.15±6.37**



(a) OA=48.46% (b) OA=47.27% (c) OA=43.86% (d) OA=40.28% (e) OA=53.78% (f) OA=50.79% (g) OA=68.33% (h) OA=82.60%

Fig. 9. Classification maps of all studied methods on the Yunyuan dataset. (a) SVM [10]. (b) PCA-EPFs [20]. (c) MSTV [25]. (d) PKCRCAWG [51]. (e) GTR [52]. (f) RPNet [53]. (g) SCMK [40]. (h) Our method.

TABLE III

CLASSIFICATION PERFORMANCE OF ALL STUDIED APPROACHES FOR YUNYUAN DATASET WITH SINGLE SAMPLE, INCLUDING SVM [10], PCA-EPF [20], MSTV [25], PKCRCAWG [51], GTR [52], RPNET [53], SCMK [40], AND OUR METHOD

Class name	Classification accuracies of different approaches (in %)							
	SVM	PCA-EPFs	MSTV	PKCRCAWG	GTR	RPNet	SCMK	Our method
Tobacco	35.85±17.41	63.02±18.89	**79.43±17.47**	35.74±16.97	22.37±7.72	46.87±26.71	66.51±18.60	41.08±25.72
Corn	62.68±15.59	48.89±12.29	44.19±9.95	39.10±27.85	51.39±18.62	52.51±11.04	63.66±9.37	**94.99±7.26**
Paddy	47.70±25.34	44.26±18.50	55.20±14.14	40.49±25.35	61.80±15.62	38.08±13.55	64.52±17.31	**99.84±0.20**
Fallow field	95.16±3.44	82.50±7.94	86.05±8.14	84.74±15.38	87.72±6.08	47.81±19.79	71.79±14.18	**99.89±0.05**
Mixed Weed	15.74±5.55	24.91±12.85	21.18±5.65	9.52±5.28	16.83±4.07	**85.50±18.94**	67.64±19.80	76.84±25.77
Chinese cabbage	10.40±4.97	42.03±29.14	69.34±16.38	9.07±5.81	18.01±11.79	**77.58±18.92**	69.94±17.13	23.19±24.70
OA	48.46±16.58	47.27±8.24	43.86±5.76	40.28±12.75	53.78±8.61	50.79±10.04	68.33±7.37	**82.60±6.53**
AA	44.59±6.95	50.94±6.10	59.23±5.76	36.44±7.35	43.02±6.56	58.06±9.80	67.34±4.91	**72.64±8.83**
Kappa	34.75±15.09	29.56±8.49	28.06±6.26	24.07±8.65	37.69±9.40	35.95±11.15	56.59±8.50	**74.95±9.00**

a lot of noise labels. Furthermore, Table III reports the objective indexes of classification performance. As presented in Table III, the values of OA, AA, and KA obtained by our method are significantly higher than that of the compared method. Specifically, OA is 34.14% higher than the original SVM, which is also used in the proposed method. The greater improvement demonstrates that our training sample expansion and optimization approaches are effective.

E. Classification Results on Limited Training Samples

In this section, different numbers of training samples are utilized to evaluate the efficiency of the proposed method with limited samples. All studied methods are also considered for comparative purposes. The obtained objective indexes as a function of the number of training samples are shown in Figs. 10–12, which are estimated on the three datasets.

According to Figs. 10–12, several conclusions can be made. First of all, the proposed method acquires the best results in all cases, which illustrates the advantage of the polygon structural in classification task. Another observation is that with the amount of initial training samples increasing, the performance of all methods is improved monotonically in most cases. Comparing with other methods, the increasing rate of our method is slower, which indicates that the sample expansion is beneficial and plays an important role in our

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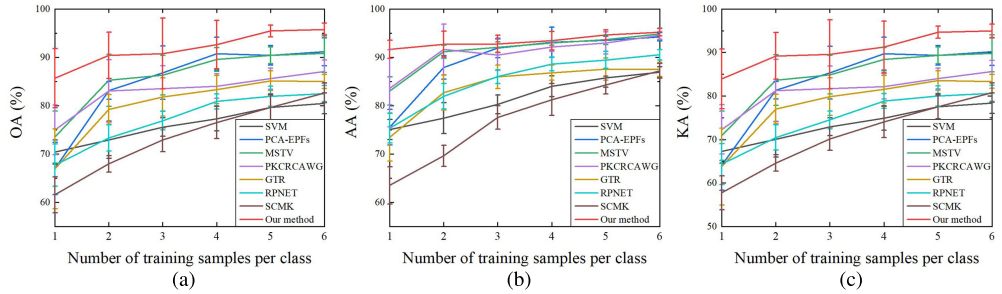

Fig. 10. Classification accuracy of Salinas when the number of training samples per class is increased from 1 to 6. (a) OA. (b) AA. (c) KA.

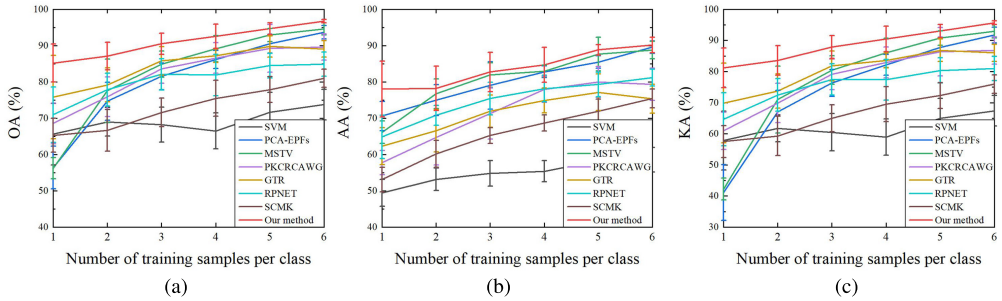

Fig. 11. Classification accuracy of WHU-Hi-LongKou when the number of training samples per class is increased from 1 to 6. (a) OA. (b) AA. (c) KA.

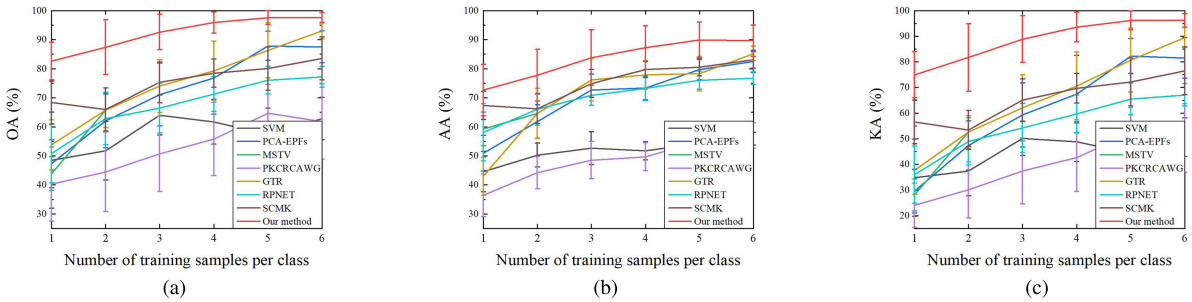

Fig. 12. Classification accuracy of Yunyuan when the number of training samples per class is increased from 1 to 6. (a) OA. (b) AA. (c) KA.

TABLE IV

CLASSIFICATION PERFORMANCE OF "POLYGON-GUIDED SPECTRAL–SPATIAL CLASSIFICATION" FOR ALL DATASETS. THE STANDARD DEVIATION (IN THE PARENTHESES) OF THE ACCURACIES OBTAINED IN REPEATED EXPERIMENTS IS ALSO REPORTED

Data sets	Classification accuracies before and after "Polygon Guided Spectral-spatial Classification".					
	OA (original)	OA (optimized)	AA (original)	AA (optimized)	Kappa (original)	Kappa (optimized)
Salinas	76.42(2.75)	**85.67(6.11)**	86.50(1.87)	**91.64(1.88)**	73.57(3.15)	**83.93(6.89)**
WHU-Hi-LongKou	82.36(4.52)	**85.20(5.24)**	74.59(6.84)	**78.06(7.69)**	77.69(5.34)	**81.15(6.37)**
Yunyuan	74.06(7.01)	**82.60(6.53)**	63.79(5.95)	**72.64(8.83)**	64.01(8.52)	**74.95(9.00)**

method. Furthermore, all these experiments demonstrate that the proposed polygon structural guided method can produce satisfactory classification accuracies with very limited training samples since spatial–spectral information is utilized to enrich the training set.

F. Ablation Experiment

1) *Effect of "Polygon-Guided Spectral–Spatial Classification":*
In the proposed framework, the third step "Polygon-Guided Spectral–Spatial Classification" which is designed to optimize the original classification map obtained by SVM. To evaluate the effect of this step, the objective indexes before and after polygon-guided optimization are shown in Table IV. It is obvious that through the optimization process, the accuracy has been greatly improved. Especially for Salinas data, OA is increased from 76.42 to 85.67.
2) *Effect of the Proposed Framework With Deep Learning Model:*
To evaluate the applicability of the proposed framework, other classifier is embedding in this framework. Specifically, the CNN model provided in [54] is utilized

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TABLE V

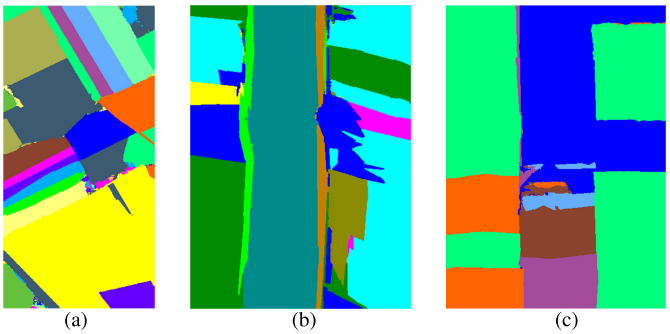
CLASSIFICATION PERFORMANCE OF THE PROPOSED FRAMEWORK WITH CNN MODEL. THE STANDARD DEVIATION (IN THE PARENTHESES) OF THE ACCURACIES OBTAINED IN REPEATED EXPERIMENTS IS ALSO REPORTED

Data sets	Classification accuracies before and after "Polygon Guided Spectral-spatial Classification".					
	OA (original)	OA (optimized)	AA (original)	AA (optimized)	Kappa (original)	Kappa (optimized)
Salinas	88.28(5.77)	**91.01(7.29)**	91.44(2.93)	**92.53(3.52)**	86.80(6.62)	**89.85(8.31)**
WHU-Hi-LongKou	84.28(2.89)	**86.02(3.01)**	75.59(4.41)	**78.73(4.44)**	79.87(3.63)	**82.06(3.77)**
Yunyuan	90.50(9.39)	**92.74(7.78)**	87.15(8.23)	**90.48(8.51)**	86.33(12.94)	**89.32(11.20)**

TABLE VI

COMPUTING TIME OF DIFFERENT APPROACHES, INCLUDING SVM [10], PCA-EPF [20], MSTV [25], PKCRCAWG [51], GTR [52], RPNET [53], SCMK [40], AND OUR FRAMEWORK WITH DIFFERENT CLASSIFIERS (SVM AND CNN)

Data sets	Running time of all studied approaches (in seconds)								
	SVM	PCA-EPFs	MSTV	PKCRCAWG	GTR	RPNet	SCMK	Our method(SVM)	Our method(CNN)
Salinas	1.38	2.80	9.93	1.39	6.86	2.95	**1.36**	1023.63	417.50/0.40
WHU-Hi-LongKou	**1.82**	6.83	23.16	3.89	22.12	5.46	2.67	140.65	728.49/1.55
Yunyuan	**1.52**	5.96	19.92	3.16	13.34	4.08	1.71	72.96	904.64/1.12



(a) (b) (c)

Fig. 13. Classification maps of our framework with CNN model and single samples on all datasets. (a) Salinas (OA = 91.01%). (b) WHU-Hi-LongKou (OA = 86.02%). (c) Yuanyuan (OA = 92.74%).

to classify the three HSIs with single sample. In this experiment, all parameters in the framework remain unchanged except that the SVM is replaced by CNN. Besides, the parameter setting of CNN is the same with [54]. The input size of image is $16 \times 16 \times 20$. Three convolutional layers are utilized and the sizes of the three convolutional filters are $3 \times 3 \times 32$, $3 \times 3 \times 164$, and $3 \times 3 \times 128$, respectively. The batch size is set to 128, and the learning rate and the number of training epochs are 0.001 and 200, respectively. Classification results are presented in Table V and Fig. 13. It can be seen that the proposed framework is able to get satisfactory results with deep learning method.

G. Computing Cost

In this article, all experiments are implemented on the 3.7-GHz CPU and 32-GB memory computer with MATLAB 2018b. Table VI shows the processing time of all studied methods with single training sample per class. It should be noted that the last column in Table VI includes training time (left) and testing time (right). According to Table VI, it can be observed that the time of all methods is acceptable. Relatively speaking, the processing time of our method is long since lots of samples are used to train the CNN model and SVM classifier with cross validation. If fix the parameters of SVM classifier, computing time will reduce significantly. SVM classifier can also be replaced by a faster classifier, i.e., MLR, to reduce the computing cost in the future.

V. CONCLUSION

A novel framework for HSIs classification with single sample is proposed in this work, which considered the sample expansion and postprocessing spatial optimization. At first, the spectral dimension of HSI is reduced with MNF to decrease the influence of noise. Then, KIPPI is performed on the first three components with image enhancement to obtain the polygonal partition map. Next, a spectral similarity-based sample expansion method is designed to get more effective information for training classifier, which also adopts polygon structure as spatial constraint. Finally, the initial classification map is further optimized by polygon structure to obtain the final classification result.

Experiments conducted on three datasets demonstrate that the polygon structure is able to improve the classification accuracy, and the proposed framework can solve small training sample size problems in man-made environments. Moreover, how to construct discriminant features to improve classification performance and reduce the calculation expense will be investigated in the future and testing the proposed method in more scenarios.

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